- The increase in temperature is sustained for a period of eighteen months to Two years.
- The temperature increase is up to 30 m beneath the ocean surface.
- When there is a modified vertical air circulation above the Pacific Ocean

Global influence of El Nino

El Nino effect is experienced at Global level. The change in air circulation affects the economy of different countries also. Global weather patterns are altered to such an extent that they affect eco system, agriculture, tropical cyclone, drought, forest fire, floods and flood related health hazards. El Nino influences the jet streams. Due to this phenomenon California experiences heavy rainfall, northern Europe experiences dry winter, Southern Europe experiences mild wet winters, there are less number of cyclones in Sea of Japan, and heavy rain in East Africa. South East Asia experiences severe drought and forest fire. Peru in South America receives heavy rainfall during El Nino.

Increase of temperature in the east Pacific Ocean is correlated with normal monsoon conditions in India while the increase of temperature in the central Pacific has high correlation with drought conditions in India. When temperature increases further to the west it suppresses the Indian Monsoon.



International **R**esearch **I**nstitute **C**limate **P**rediction **C**entre predicts and forecasts El Nino occurrences. Scientists are in the opinion that El Nino can cause Global Warming and it also increases the frequency of El Nino occurrence.

Normal Situation	El Nino Situation
➢ Near equator the water of the Pacific Ocean is warmer in the western side and cooler in the eastern side due to upwelling of the cold current. ➢ Air (Walker) circulation is dominant in the western Pacific Ocean. The air ascends in the western side and descends over the cooler eastern side ➢ Heavy rain is experienced in the western warmer region and dry conditions prevail in the cooler region. ➢ The Southeast Asia and Australia receive heavy rain on normal years. ➢ West coast of South America experiences dry weather.	➢ Near equator the warm water in the Pacific Ocean extends from western side to eastern side suppressing the upwelling of the cold water. ➢ Air (Walker) circulation is dominant in the eastern part of Pacific Ocean. The air ascends in the warm eastern Pacific Ocean. ➢ Heavy rain is experienced in the eastern warmer region and dry condition prevails in the western part. ➢ Southeast Asia and Australia experience dry weather conditions. ➢ West coast of South America receives heavy rainfall.

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